NITISH VATTIKUTI

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Education

MVGR College of Engineering — Expected May 2027
Bachelor of Technology in Computer Science and Engineering (CGPA: 7.59) — *VIZIANAGARAM, AP*

- **Relevant Coursework:** DSA, Operating System, DBMS, Computer Networks, OOP(C++/Java), AI-ML.

Experience

VANTIRIS TECHNOLOGIES LLP — Remote, India
Software Engineering Intern — *February 2026 – Present*

- **Web Optimization & Scaling:** Developed and deployed responsive frontend architectures for international logistics platforms (APET Logistics, Marine Commercial Ltd), expanding deployment infrastructure across optimized Netlify environments.
- **UI/UX Redesign:** Spearheaded a comprehensive user interface overhaul of legacy, raw HTML client applications into responsive, multi-viewport web structures engineered for cross-device compatibility.
- **Technical Workflows:** Leveraged Git/GitHub version control workflows to collaborate seamlessly during weekly engineering synchronization meetings, accelerating iterative feature deployments.
- **Project Architecture:** Documented comprehensive project lifecycles, backend logic structures, and technical architectural tradeoffs to establish long-term engineering documentation guidelines.

Projects

AUTONOMOUS UAV SIL SIMULATION | *Python, Socket Programming, Docker, Avionics* | *(Github Repository)* | *(Render Link)*

- Designed a 3D Software-in-the-Loop (SIL) UAV guidance simulation with dynamic Sense-and-Avoid (SAA) capabilities.
- Optimized 3D path planners using Weighted A* and RRT*, reducing planning latencies to under 80ms.
- Built a dual-process TCP/IP socket link to stream real-time avionics telemetry from an emulated CPU to a Web GCS.
- Profiled algorithm execution under synthetic CPU stress using psutil to validate strict hardware SWaP constraints.

SPECTRAFUSE | *TensorFlow JS, COCO-SSD, HTML5 Canvas API, Zustand* | *(Github Repository)* | *(Netlify Link)*

- Created a high-performance multi-spectral image fusion platform emulating DRDO CABS aerial surveillance requirements to merge Visible, NIR, and TIR aerial imagery.
- Implemented client-side DWT, PCA, and IHS pixel-level mathematical transforms to generate high-fidelity composite surveillance data.
- Ensured absolute data isolation for classified defense assets by building a 100% local browser pipeline with zero server overhead. Integrated browser-based TensorFlow models for tactical object detection alongside real-time image quality evaluation metrics (PSNR, SSIM).

Technical Skills

Languages: Python, SQL, Java, C/C++ (OOP)
Frameworks & Libraries: NumPy, OpenCV, TensorFlow.js, HTML, CSS, JavaScript
Tools & Systems:Git/GitHub, Linux, Docker, MySQL, ServiceNow (CSA/CAD), PowerBI, Figma, MS Office 365

Certifications

ServiceNow Certified System Administrator (CSA) *(Certificate link)* – ServiceNow — 2026

ServiceNow Certified Application Developer (CAD) *(Certificate link)* – ServiceNow — 2026

HTML, CSS, and Generative AI: Speed Up Your Process *(Certificate link)* – LinkedIn — 2024

Python Essentials *(Certificate link)* *(Badge link)* – Cisco Networking Academy — 2025